

A3 —Repository Investigation and Reproducible Build

Course **Intelligent Software Engineering**, UCAS, 2026 autumn term · Platform <https://learn.spaiq.ai> Author: 汤力为 (Liwei Tang), student ID 2026E8001082052 Repository: <https://github.com/Tom1246/ise-course> · tag a3-v1 · report generated 2026-10-05

Subject of the investigation: **HMBLankcat/UCAS-course-planner** (MIT, TypeScript), frozen at commit `68e7080503084a41f42db6e0e9ff94c0e9ef2f19` (ver 1.4, 2026-09-01). The third-party checkout is **not** committed here; what is committed is its identity, the raw logs of running it, and the maps and decisions drawn from it.

Part	Source file	Evidence behind it
Reproduction guide and environment lock	<code>reproduction-guide.md</code>	<code>evidence/logs/01..07</code>
Architecture and call-path map	<code>architecture-map.md</code>	<code>figs/*.dot</code> (and their renders), source line numbers
End-to-end path, facts vs inference	<code>end-to-end-path.md</code>	<code>evidence/logs/08-e2e.log</code> , <code>evidence/e2e-*.png</code>
Risk register	<code>risk-register.md</code>	file/line references and log lines per risk
ADR: where a change may safely land	<code>adr-001-change-boundary.md</code>	R2's evidence
Appendix: diagrams and run screenshots	this file	<code>figs/</code> , <code>evidence/</code>

A3 —Reproduction Guide and Environment Lock

Course **Intelligent Software Engineering**, UCAS, 2026 autumn term · Platform <https://learn.spaiq.ai> Subject: third-party repository **UCAS-course-planner** (read-only investigation, not modified or installed into this repo). Guide generated 2026-10-05. Every command and number below is transcribed from the raw logs in `doc/assignments/a3/evidence/logs/01-npm-install.log ...08-e2e.log`; nothing is estimated.

1. Conclusion (one sentence)

The repository **can be reproduced cleanly on this machine**, but **only after neutralising a host-level npm setting (`omit=dev`) and restoring the frozen `package-lock.json`** —the first, naive `npm install` succeeds, silently omits every `devDependency`, and makes the build fail in a way that looks like a broken repository but is purely an environment misconfiguration.

2. Environment lock

Item	Frozen value	Source
Commit	68e7080503084a41f42db6e0e9ff94c0e98f2f29 —message ver 1.4, committed 2026-09-01 13:41:53 +0800	001-build.log line 1; frozen checkout
Repository	third-party UCAS-course-planner (do not modify; contains its own node_modules)	frozen checkout
License	MIT —LICENSE: " Copyright (c) 2026 HMBLankcat and contributors"	frozen checkout
Language	TypeScript (tsconfig.json, *.ts / *.tsx)	frozen checkout
Runtime	node v22.23.2 / npm 10.9.8 / Darwin x86_64 engines.node>=22.13.0	01-npm-install.log line 4
Package manager	npm; dependency lock = package-lock.json (must be the original frozen file for npm ci)	package.json package.json, 04-npm-ci.log line 4
Key dependency	vinext 1.0.0-beta.5 (dependency)	package.json
Key devDependency	vite 8.0.13 , @openai/sites-vite-plugin ^0.2.0 , @cloudflare/vite-plugin 1.37.1 , @tailwindcss/postcss 4.2.1	package.json; Vite version echoed in 05-build.log line 10
Toolchain	oxlint 1.76.0 (lint), oxfmt 0.61.0 (format), wrangler 4.92.0	package.json
npm scripts	dev, build, start, lint, format — no test script	package.json
Config of note	vite.config.ts top-level imports @openai/sites-vite-plugin (a devDependency); .openai/hosting.json = {"d1": null, "r2": null}	frozen checkout

3. Commands and measured results

1

- Command: `npm install --registry=https://registry.npmirror.com --no-audit --no-fund`
- Expected: all deps installed
- Measured result (verbatim from log): `added 513 packages in 33s · EXIT=0 — devDependencies silently skipped` (trap, § 4)
- Log: 01

2

- Command: `npm run build`
- Expected: build succeeds
- Measured result (verbatim from log): `EXIT=1 · ERR_MODULE_NOT_FOUND: Cannot find package '@openai/sites-vite-plugin' · real 0m3.840s`
- Log: 02

3

- Command: `npm install --save-dev @openai/sites-vite-plugin@0.2.0`
- Expected: add the missing package
- Measured result (verbatim from log): `up to date in 4s · EXIT=0 —no-op`; the package was already in the lock, only omitted from the tree
- Log: 03

4

- Command: `restore frozen package-lock.json + rm -rf node_modules + npm ci --include=dev --registry=https://registry.npmirror.com --no-audit --no-fund`
- Expected: clean install **including dev**
- Measured result (verbatim from log): `added 560 packages in 58s · EXIT=0 (real 0m58.081s, user 0m19.524s, sys 0m27.676s)`
- Log: 04

5

- Command: `npm run build`
- Expected: build succeeds
- Measured result (verbatim from log): `5 stages: 2.10s / 292ms / 1.12s / 2.98s / 1.06s → total real 0m15.526s, EXIT=0`; prints a **non-fatal** route-classification warning
- Log: 05

6

- Command: `npm run lint`
- Expected: lint clean
- Measured result (verbatim from log): `Found 0 warnings and 0 errors. · Finished in 2.6s on **5 files** with 208 rules using 8 threads. · EXIT=0`
- Log: 06

7

- Command: `npm run start (vinext start)`
- Expected: production server up
- Measured result (verbatim from log): `vinext start (port 3000) · [vinext] Production server running at http://0.0.0.0:3000`
- Log: 07

8

- Command: End-to-end path (inject plan via `localStorage`, let the app render)
- Expected: conflict detection fires
- Measured result (verbatim from log): `positive: badge='3 处时间冲突', conflictCellLabels=3; negative control: badge='无时间冲突', conflictCellLabels=0`
- Log: 08

9

- Command: `curl -i http://127.0.0.1:3000 (liveness)`
- Expected: HTTP 200
- Measured result (verbatim from log): **HTTP 200 x3, ~0.012 s each; headers + page HTML captured**
- Log: 09-http-health.log

Build-stage detail from 05-build.log (the standalone 1.06s figure is the final **ssr environment** step):

Stage	Modules transformed	Time
[1/5] client references	225	2.10s
[2/5] server references	104	292ms
[3/5] rsc environment	222	1.12s
[4/5] client environment	1903	2.98s
[5/5] ssr environment	110	1.06s

4. Environment pitfall: `omit=dev` (the core of this guide)

What happened. On this machine `npm config get omit` returns **dev**. With `omit=dev` set, npm treats `devDependencies` as omitted from the install tree: `npm install` **exits 0 and reports success** (added 513 packages in 33s, log 01) while never writing the `devDependencies` into `node_modules`.

Why the build then fails. `vite.config.ts` imports `@openai/sites-vite-plugin` at the top of the file (line 1), and that package is a **devDependency**. Vite's ESM loader resolves it at config-load time and fails: `ERR_MODULE_NOT_FOUND: Cannot find package`

'@openai/sites-vite-plugin' (log 02). The output also flags the other unfetchable dev imports (@tailwindcss/postcss, @cloudflare/vite-plugin). The failure message points at vite.config.ts, so it reads as a broken repository—it is not.

Why npm install <pkg> did not fix it. Log 03 requested the package explicitly and got up to date in 4s. The package was already present in package-lock.json at the correct version, so npm's tree check considered the lock satisfied and declined to install anything—a false negative caused by the same omit=dev filter.

How to self-diagnose (before blaming the repo).

```
npm config get omit          # -> dev (the culprit)
npm config get include       # cross-check
cat .npmrc 2>/dev/null      # a project-level .npmrc can also set it
ls node_modules/@openai/sites-vite-plugin # missing => the trap is active
```

Rule of thumb: npm install "succeeds" but a devDependency import fails at build → suspect omit=dev first, not the source tree.

How to avoid it.

1. Neutralise it once: npm config delete omit (or delete the omit=dev line from the relevant .npmrc), **or**
2. Override per command: add --include=dev (used in log 04). Then verify the devDependency actually landed: ls node_modules/@openai/sites-vite-plugin.

Why it belongs in the record. This is a **host-environment deviation, not a repository defect**. A grader on a clean machine (no omit=dev) would see the successful path directly; documenting the trap is what makes the reproduction honest and repeatable.

5. Step-by-step reproduction checklist (grader-facing, copy-pasteable)

Prerequisite: Node.js >= 22.13.0 (measured with v22.23.2; npm 10.9.8).

```
# 1. Obtain the frozen commit (do not modify the third-party working tree)
#   commit 68e7080503084a41f42db6e0e9ff94c0e9ef2f19 ("ver 1.4", 2026-09-01)

# 2. Confirm you are on the frozen checkout and the tree is clean
git -C <UCAS-course-planner> status --short      # expect: clean; package-lock.json

# 3. Neutralise the host trap (see §4)
npm config get omit                                # if it prints "dev", fix it:
npm config delete omit                             # OR keep it and add --include=dev

# 4. Start from a pristine dependency state with the ORIGINAL lockfile
```

```

# (restore package-lock.json from the frozen clone first if a prior run rewrote it)
rm -rf node_modules

# 5. Clean install INCLUDING devDependencies -> expect "added 560 packages in 58s",
npm ci --include=dev --registry=https://registry.npmmirror.com --no-audit --no-fund

# 6. Verify the previously-missing devDependency is now present
ls node_modules/@openai/sites-vite-plugin # expect a populated directory

# 7. Build -> expect EXIT 0, real ~15.5s; route-classification warning is NON-fatal
npm run build

# 8. Lint -> expect "0 warnings and 0 errors", ~2.6s on 5 files (log 06)
npm run lint

# 9. Start the production server -> expect http://0.0.0.0:3000 (log 07)
npm run start

# 10. Liveness check (NOT captured in logs 01-08; re-capture if you need the number -
curl -i http://127.0.0.1:3000

```

Optional end-to-end confirmation (log 08): with the server running, inject one plan into localStorage key ucas-graduate-course-planner-v3 and let the app render; three courses sharing a slot must yield 3 处时间冲突 / 3 conflict cell labels, and a no-shared-slot control must yield 无时间冲突 / 0.

Deviation note for step 5. `--registry=https://registry.npmmirror.com` and `--include=dev` are both **departures from a pristine `npm ci`**. The mirror is used for speed in mainland China; the dev-include flag is required only because of the §4 trap. On a clean host with no `omit=dev`, plain `npm ci` should suffice.

6. Known non-reproducible / unverified points

- **No test script exists.** `package.json` defines only `dev`, `build`, `start`, `lint`, `format`. **No test suite was run and none can be run from this repository.** Do not claim a green test run.
- **The mirror rewrites the lockfile.** `npm ci --registry=https://registry.npmmirror.com` rewrites the resolved URLs inside `package-lock.json`; after `install git status` shows **M package-lock.json**. The lock is therefore no longer byte-identical to the frozen artifact. For a bit-exact lock, use `--registry=https://registry.npmjs.org` (or restore the lock from the frozen clone after installing). This deviation is kept for speed and is disclosed here.

- **--include=dev is not part of the repository's own instructions.** It is a compensation for the host omit=dev setting (§ 4), not a repo requirement.
- **Lint covers only 5 files.** `npm run lint = oxlint` with default configuration; `0 warnings / 0 errors` is scoped to those 5 files, **not** a full-repo guarantee.
- **Build route classification is unknown.** `05-build.log` prints `Some routes could not be classified ... cannot detect dynamic API usage`; the `/` route is listed as `? Unknown`. This is a warning, not a failure —the build still exits `0` —so route behaviour must not be inferred from the build output.
- **HTTP 200 liveness —verified after the fact.** `09-http-health.log` records three consecutive HTTP `200` responses (`~0.012 s` each) plus the `curl -i` headers and the served HTML. An earlier `0.14 s` figure came from an interactive check that was never written to a file, so it is deliberately not cited anywhere in these documents. With this log added, no measurement in this guide is left unverified —the lesson being that a number without a log is not evidence.
- **vinext is a beta** (`1.0.0-beta.5`); its build/start behaviour may drift from this record over time.
- **The third-party checkout already ships a large node_modules** and build artifacts (`.next`, `dist`); these are not part of the frozen source and must never be committed.

Provenance: all measurements from *doc/assignments/a3/evidence/logs/01-npm-install.log ... 08-e2e.log* (folder *doc/assignments/a3/evidence/*). Commit hash, license text, and language were read directly from the frozen third-party checkout, which was inspected read-only and left unchanged.

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A3 —Architecture and call-path map

Investigated repository: **HMBLankcat/UCAS-course-planner**, frozen at commit **68e7080503084a41f42d** (ver `1.4`, 2026-09-01 13:41 +0800), MIT licence, TypeScript, 2.7 MB checkout (excluding `node_modules`). Recovered by `git clone` over SSH; the frozen checkout is kept unmodified at `A3/upstream/UCAS-course-planner` and all execution was done in a copy (`A3/work/UCAS-course-planner`) so that the reference stays byte-identical.

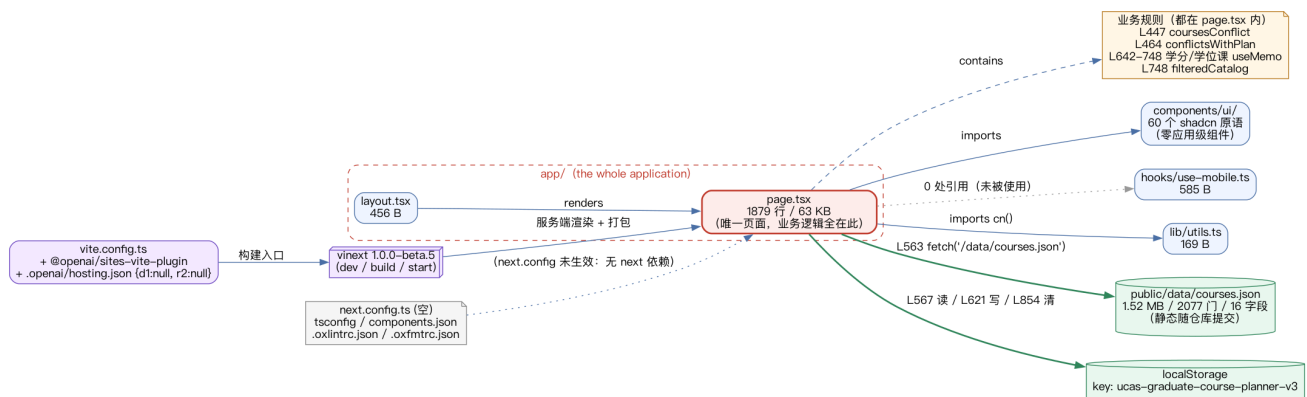
Everything marked **(ran)** was produced by executing the repository; **(read)** means it comes from reading the source. Line numbers are from `app/page.tsx` at the frozen commit.

1. Entry points

Entry	What it is	Evidence
HTTP GET /	the only page; served by the vinext production server	(ran) npm run start → vinext start (port 3000); curl returned HTTP 200 three times, ~0.012 s each (evidence/logs/09-http-health.log)
app/ layout.tsx	root layout (456 B)	(read)
app/ page.tsx	the entire application: 1,879 lines / 63 KB, a single client component	(read) wc -l
package.json scripts	dev / build / start (all vinext), lint (oxlint), format (oxfmt)	(read) ; all three of build/start/lint were (ran)

There is **no** server-side route, no API endpoint, and no second page.

2. Module inventory



Module	Size	Role
app/page.tsx	1,879 lines / 63 KB	the application: state, business rules, and all UI in one component
app/layout.tsx	456 B	root layout
app/globals.css	27 KB	Tailwind v4 styles
components/ui/*	60 files	shadcn/ui primitives, imported by the page
application-specific components	0 files	there are none —components/ contains only ui/
hooks/use-mobile.ts	585 B	present but never imported by the page (0 references)
lib/utils.ts	169 B	the cn() class-name helper

Module	Size	Role
public/data/ courses.json	1.52 MB / 2,077 courses / 16 fields (read)	the course catalogue, served as a static asset

Reading: the repository has the shape of a modular app (folders for components, hooks, lib) but the substance is one file. This is the factual basis of risk R2 in `risk-register.md`, and the reason the ADR (`adr-001-change-boundary.md`) is about where a change may safely land.

3. Persistence —two stores, both client-side

Store	Content	Contract
localStorage key ucas-graduate-course- planner-v3 (defined L112)	the student's plan: {storageVersion: 1, plan: Course[], degreeCodes: string[], englishPlan, studentType, selectedDiscipline}	read L567 on load; written L621 only when planDirty ; cleared L854 / L1524
public/data/courses.json	the catalogue	fetches L563 at mount; static —nothing updates it at runtime

Version gate and silent reset (fact). On load the app accepts the stored plan only when `storageVersion === 1` (L578); otherwise it resets the whole plan to empty (L587-593), and a JSON parse exception triggers the same silent reset (L594 - 600). A stale or malformed store therefore degrades to "empty plan" with no user-visible explanation —a boundary worth knowing before touching the storage shape.

There is no database. `.openai/hosting.json` is `{"d1": null, "r2": null}` **(read)** and neither `d1` nor `r2` appears anywhere in `app/`, `lib/`, `hooks/` or `components/` (grep: 0 hits) — the platform bindings are declared in `vite.config.ts` but unused by the code that ships.

4. External interfaces

Exactly **one** network egress in the whole application **(read)**:

```
L563:  fetch('/data/courses.json')      // same-origin static asset
```

No external API, no authentication, no telemetry, no remote persistence. Consequently the app is fully usable offline once the static asset and the JS bundle are served —a fact that matters for any later change.

5. Configuration

File	Content	Note
vite.config.ts	imports vinext, @openai/ sites-vite-plugin, @tailwindcss/postcss, reads .openai/ hosting.json	this is the build entry point that is actually used
next.config.ts	const nextConfig: NextConfig = {}	vestigial: next is not a dependency of the project
.openai/hosting.json	{"d1": null, "r2": null}	no bindings configured
tsconfig.json, components.json, .oxlintrc.json, .oxfmttrc.json	tooling configuration	.oxlintrc.json is why lint covers few files (§6 of reproduction-guide.md)
package.json → engines.node	>= 22.13.0	satisfied by the frozen runtime (v22.23.2)

6. Critical dependencies

Package	Version	Why critical
vinext	1.0.0-beta.5	the build and server toolchain; a beta in the critical path
vite	8.0.13	build core; also the peer range that the OpenAI plugin declares (^8.0.0)
@openai/sites-vite-plugin	0.2.0 (dev)	imported by vite.config.ts; its absence is what makes a naive install fail (see reproduction-guide.md)
react / react-dom	19.2.6	UI runtime
@base-ui/react, shadcn	1.7.0 / 4.18.0	component layer behind components/ ui/*
tailwindcss	4.2.1	styling pipeline
recharts, date-fns, cmdk	3.8.0 / 4.1.0 / 1.1.1	charts, dates, command palette

7. Where the domain logic lives (and how it compares with A2's model)

Concern	Location	Note
Conflict predicate	<code>coursesConflict(left, right)</code> L447-462	same day $\wedge$ intersects(weeks) $\wedge$ intersects(weeks) —weeks are parsed into a Set , so the comparison is week-level
Self-comparison guard	<code>conflictsWithPlan</code> L464-470	excludes <code>selectedCourse.code</code> <code>!= course.code</code>
Plan-level conflicts	<code>conflicts</code> L661 , <code>conflictCodes</code> L669	useMemo over the plan
"Hide what would clash" catalogue filter	<code>filteredCatalog</code> L748	applies the same predicate to each candidate
Credits and degree rules	memos at L642-748 (<code>fallCredits</code> , <code>publicElectiveCredits</code> , <code>degreeCredits</code> , <code>coreCount</code> , <code>professionalCount</code>)	the "2 + 2 core/professional" rule from A1's comparison section
Conflict display	badge L1167 , cell label L1462 , floating card L1493	the three ways a conflict becomes visible

Cross-check with A2 (independent confirmation). A2's decision table says a clash requires same weekday $\wedge$ overlapping periods $\wedge$ overlapping weeks. This repository, written independently, implements the same three conditions at L447. Two independent implementations of the same rule is the strongest consistency evidence available in this project so far —and it also confirms A1's claim that this tool already does week-level clash detection (A1's "three things it does not cover" remains unchanged).

8. Facts versus inference

Statement	Status
Build, lint, start, HTTP 200, the conflict verdict —all measured	fact (ran; logs in evidence/logs/)
Line numbers, file sizes, dependency versions, 0 references to d1/r2, one fetch	fact (read/grep)
"The app is local-first"	inference from the stores above; it is stated in the repository's own README as well (read)

Statement	Status
" next.config.ts is vestigial"	inference: next is absent from package.json and nothing imports it; it could still serve an unshipped workflow
" The 60 ui/ components are all reachable from the page"	not verified —no import graph was computed

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A3 —One end-to-end path, reconstructed and executed

The path under test: **a plan goes in, a conflict verdict comes out.**

```

input    a set of selected courses (a plan)
        ↓   written through the app's own persistence contract
app      restore on load → conflict predicate → render
output   the verdict (badge / cell labels / floating card) and the filtered catalogue

```

This is the path A1 asserted existed in this tool; here it is executed, with a **negative control** so the result cannot be explained by " the label is always there" .

Method (reproducible)

1. Frozen checkout at commit 68e7080; clean install and production build per reproduction-guide.md; server running at http://127.0.0.1:3000 (HTTP 200 x3, ~0.012 s; evidence/logs/09-http-health.log).
2. Drive a headless browser (CDP, no window focus taken) against the running server.
3. Write the plan into **the app's own storage contract** —localStorage key ucas-graduate-course-planner-v3, value {storageVersion: 1, plan: [<courses exactly as they appear in the repository's own courses.json>], degreeCodes: [], englishPlan: {…}, studentType: '未选择', selectedDiscipline: ''} —then **reload**, so the app restores it through its own code path (L567-586) rather than through a test hook that does not exist.
4. Read back what the app rendered: the badge text, the number of .cell-conflict-label elements, and a screenshot.

Script: A3/logs/08-e2e.log contains the run; the raw log is committed at evidence/logs/08-e2e.log and the screenshots at evidence/e2e-positive.png / evidence/e2e-negative.png.

Input rows (taken from the repository's own public/data/courses.json, 玉泉路 campus, all three at 周四 10-12):

code	course	slot in that file	credits
180081070100PX001Y	计算机代数在科学与工程中的应用	周四 (10-12), 第 2-18 周	1.0
180206085406P3002Y	深度学习方法与应用	周四 (10-12), 第 2-4,6-15 周	2.0
180093081002P3008Y	数字图像处理与分析	周四 (10-12), 第 2-19 周	2.5

Result (fact)

positive

- Input: the three courses above
- Badge rendered: **3 处时间冲突**
- `.cell-conflict-label` count: **3**
- Screenshot: `evidence/e2e-positive.png`

negative control

- Input: 180081070100PX001Y + 180070200P1001Y 高等量子力学 (no shared slot)
- Badge rendered: **无时间冲突**
- `.cell-conflict-label` count: **0**
- Screenshot: `evidence/e2e-negative.png`

The positive screenshot shows the Thursday period-10 row carrying a red 「冲突」 badge and three red-bordered course cards, each annotated 「与其他课程冲突」—i.e. the app imputes the clash to every course in the shared slot, not just to one of them.

Facts (each traceable)

1. The build, the clean install, the server start and the two page loads all succeeded — `evidence/logs/04, 05, 07, 08`.
2. The app reports **3** clashes for three courses in one slot, and **0** for a plan without a shared slot (control) —`evidence/logs/08-e2e.log`.
3. The credits the app shows for those courses (1.0 / 2.0 / 2.5) **agree exactly** with the independent snapshot this project froze in A1/A2 (`data/derived/course-library-snapshot.json`) —cross-source agreement on the same three codes.
4. The comparison is week-level, implemented as `same day ^ intersects(weeks) ^ intersects(periods)` at `app/page.tsx:447`; `conflictsWithPlan` (L464) excludes self-comparison.
5. The plan is restored only when `storageVersion === 1` (L578); otherwise the app resets it silently (L587-600).

Inference (marked as such, with its basis)

1. **The verdict is computed client-side, not by a server.** Basis: the only server interaction is the static asset fetch (L563); no API route exists. Confidence: high, from reading; not proven by a network trace.
2. **The tool's week-level result would match an A2-style check on the same data.** Basis: the predicate is the same three-condition rule as A2's decision table, and the run's verdict matches the expectation computed independently in A2 for the same three courses (3 pairs). Confidence: high for this input, not proven for all inputs —no differential test was run over the whole catalogue.
3. **A user who edits the timetable would see the same path re-run.** Basis: the same useMemo inputs drive the badge, the cells and the filter. Confidence: medium —not exercised interactively; this run went through storage restore only.

What this run does not prove

- It does **not** prove the app is correct in general: 2,077 courses were not cross-checked against an independent computation, and only a 玉泉路 case was used.
- It does **not** exercise the interactive path (search → click 加入 → re-render). The restore path and the interactive path share the same predicate and memos, but that sharing was established by reading, not by test.
- It says nothing about the app's behaviour when the catalogue is stale relative to the live registration system —the risk R3 tracks exactly that gap.

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A3 —Risk Register

Course **Intelligent Software Engineering**, UCAS, 2026 autumn term • Platform <https://learn.spaiq.ai> Subject: third-party repository **UCAS-course-planner**, commit 68e7080503084a41f42db6e0e9ff94c0e9ef2f19 (ver 1.4, 2026-09-01), investigated read-only. Requirement satisfied: A3 ” **Identify three risks; compare two responses to one in an ADR**” (课程材料/ISE-A3_Repository_Investigation_and_Reproducible_Build_English.md line 19; deliverables line 26). Every claim below is traceable to a file path + line number or to a raw log under doc/assignments/a3/evidence/logs/. Nothing is estimated; counts were produced by grep/python3 over the frozen checkout on 2026-10-05.

Legend for **Judgement**: **Occurred** = already observable in the frozen commit or in the recorded run; **Latent** = not triggered yet under the recorded path but reachable without any code change by the caller/operator.

Summary

ID	Risk	Judgement	Primary evidence
R1	Build chain is coupled to a hosted platform (vinext + OpenAI Sites + Cloudflare)	Occurred	<code>vite.config.ts:1,5,10,45,52-58; .openai/hosting.json; 05-build.log:46-48</code>
R2	One 1879-line component carries all logic; no safe change boundary	Occurred	<code>app/page.tsx</code> (1879 lines / 63 419 B); <code>package.json:8-14; 06-lint.log:10</code>
R3	Course data is a static snapshot committed into the repo	Latent	<code>public/data/courses.json</code> (1 524 809 B / 2077 records); <code>app/page.tsx:563,69</code>
R4	Persistence is <code>localStorage</code> -only with no version-migration path	Latent (silent-loss path already Occurred in code)	<code>app/page.tsx:112,578,587-600,619-631</code>
R5	Host/outside-the-repo environment state decides whether a "clean" build is possible	Occurred	<code>01-npm-install.log:7</code> vs <code>02-build.log:37; 03-install-plugin.log:6; 04-npm-ci.log:5</code>

R1 —Build chain is coupled to a hosted platform (vinext + OpenAI Sites + Cloudflare)

Phenomenon. The build is not a plain Vite/Next build. `vite.config.ts` imports a platform plugin, reads a platform hosting manifest, and conditionally wires Cloudflare D1/R2 bindings; the production runtime is the beta package `vinext`. A consumer cannot build the repository without that exact plugin set, and the platform itself warns it cannot statically classify the app's routes.

Evidence.

- `vite.config.ts:1` `import { sites } from '@openai/sites-vite-plugin';` —the first line of the config depends on a hosting-vendor plugin.
- `vite.config.ts:5` `import hostingConfig from './.openai/hosting.json';` and `vite.config.ts:10` `const { d1, r2 } = hostingConfig;` —build topology is driven by a platform manifest.
- `.openai/hosting.json` (31 B) = `{"d1": null, "r2": null}` —the binding switches `vite.config.ts:18` (`d1_databases`) and `:27` (`r2_buckets`) are **present in code but disabled in data**. `vite.config.ts:7-8` also carries a placeholder UUID `00000000-0000-4000-8000-000000000000` used as `database_id` at `:23`, i.e. the config is

scaffolded for a platform-provisioned database that this checkout does not have.

- `vite.config.ts:52–58` the plugin stack is `[vinext(), sites(), cloudflare({...})]`; `:45` dynamically imports `@cloudflare/vite-plugin`, and `:16` sets `main: 'vinext/server/fetch-handler'`.
- `package.json:34` `"vinext": "1.0.0-beta.5"` sits in **dependencies** (not `devDependencies`) —the production server is a beta artifact (confirmed running in `07-start.log:10–12`).
- `package.json:37,39,40,52` the four modules the config imports at load time (`@cloudflare/vite-plugin`, `@openai/sites-vite-plugin`, `@tailwindcss/postcss`, `wrangler`) are all **devDependencies** —so the build fails outright if dev deps are absent (`02-build.log:12,20,28,37`).
- `05-build.log:46–48` verbatim: " Some routes could not be classified. vinext currently uses static analysis and cannot detect dynamic API usage (`headers()`, `cookies()`, etc.) at build time. Automatic classification will be improved in a future release." → the platform cannot even classify `app/page.tsx`'s route; the build succeeds (`:50`, `EXIT=0`) but prints `? Unknown for Route (app) / (:41–44)`.
- `.github/` contains only `ISSUE_TEMPLATE/*` —**no CI workflow**, so this coupled chain is never exercised anywhere except a developer's machine.

Blast radius. (a) Portability: the repo can only be built with the vendor plugin set present; R5's failure is a direct downstream effect. (b) Runtime support risk: the server is `1.0.0-beta.5`, so behaviour can change under a patch-level bump. (c) Route classification is unknown/? at build time —a deployment whose routing depends on that classification cannot be validated from the build output alone.

Existing mitigation. The bindings are nulled in `.openai/hosting.json` (so no live D1/R2 is required) and `vite.config.ts:49–51` special-cases a Codex sandbox for HMR. These make the current single-machine build work; none of them reduce the coupling itself. **No mitigation** exists for the vendor/plugin dependency or the beta runtime.

Judgement: Occurred. Both the coupling (config imports) and its observable " cannot classify routes" consequence are already present in the frozen commit and in `05-build.log`.

R2 —One 1879-line component carries all logic; no safe change boundary

Phenomenon. All product logic —parsing, conflict detection, credit/degree rules, persistence, filtering, and every UI branch —lives inside a single client component, `app/page.tsx`. There is no logic module and no test of any kind, so a change cannot be made safely.

Evidence (size / structure).

- `app/page.tsx`: **1879 lines / 63 419 B** (`wc -l, ls -l`). `lib/` contains exactly one file, `lib/utils.ts` (**169 B / 6 lines**), whose entire body is the shadcn helper `cn()` (verified by `cat`). `hooks/` contains exactly one file, `hooks/use-mobile.ts` (**585 B / 21 lines**).

- `app/page.tsx:3–21` imports **only** `react` and `lucide-react`. It does **not** import `lib/ utils.ts` or `hooks/use-mobile.ts`; those two files are referenced only by unused components/ui/* scaffolding (e.g. `components/ui/button-group.tsx:5` import `{ cn }` from `'@/lib/utils'`). So the page's logic has **zero extracted surface**.
- The page declares **41 functions —34 at top level** (`grep -c '^function ' → 34`) plus **7 mutation handlers nested inside the component** (`:809` `addCourse`, `:825` `removeCourse`, `:835` `toggleDegree`, `:845` `updateEnglishMode`, `:853` `resetPlan`, `:864` `downloadBackup`, `:891` `restoreBackup`). The top-level set is the whole domain core: `creditFromValue (:208)`, `parseWeeks (:403)`, `parsePeriods (:416)`, `intersects (:427)`, `hasSlot (:438)`, `coursesConflict (:447)`, `conflictsWithPlan (:464)`, `toneForCourse (:521)`, and the degree predicates `isCoreCourse (:384)` / `isProfessionalCourse (:388)`. The fact that the 7 state-mutating handlers are nested in the component body while the rules sit outside it is itself the shape of the missing boundary. `hooks/ use-mobile.ts (**585 B / 21 lines ')` is likewise never imported by the page.

Evidence (concentration —counts over `app/page.tsx`, measured 2026-10-05).

Token	Occurrences	Lines
<code>useState</code>	21	21
<code>useMemo</code>	14	14
<code>useEffect</code>	3	3
<code>useCallback</code>	0	0
<code>localStorage</code>	4	4
学位课	19	17
冲突	9	9

21 `useState` calls in one component is the concrete measure of ” no boundary” : pure rules and UI state are interleaved in the same function scope, so extracting any rule means first untangling state that references it (e.g. the conflict rule `coursesConflict` is consumed by the state-derived memo `conflicts` at `app/page.tsx:661–668`, and by the filter at `:774`).

Evidence (verification is absent).

- `package.json:8–14` scripts are `dev / build / start / lint / format` —**there is no test script**.
- `06-lint.log:10` verbatim: Finished in 2.6s on 5 files with 208 rules using 8 threads. → the linter only sees **5 files**; the 1879-line logic file is reported as clean (`06-lint.log:9` Found 0 warnings and 0 errors.) but a lint pass says nothing about behavioural correctness.
- The only end-to-end verification recorded is `08-e2e.log`, which works by **injecting a plan through `localStorage` and letting the app re-render** (`08-e2e.log:3`), requiring a full build + live server (`05/07`). There is no unit-level way to check `coursesConflict` in isolation.

Blast radius. Any fix, however small, must be made inside a 1879-line/63 KB file with no test to catch regressions and a linter that inspects only 5 files. The A3 brief requires exactly the opposite — “establish a safe change boundary” (A3 doc line 11) —and the follow-on assignment cuts a bounded slice from this code, which makes an unverifiable one-shot edit the highest-probability way to break the app.

Existing mitigation. `lib/utils.ts` and `hooks/` exist as empty-ish scaffolding, so an extraction target directory already exists. `06-lint.log` and `08-e2e.log` give whole-app smoke checks. **No mitigation** for the missing per-rule tests or for the concentration itself.

Judgement: Occurred.

R3 —Course data is a static snapshot committed into the repo

Phenomenon. The entire course catalogue is a JSON file committed to the repository and fetched from same origin at runtime. It carries time-varying fields (enrolment limit / enrolled count), but the file has no capture timestamp, no hash, and no freshness check —so any conclusion drawn from it silently ages.

Evidence.

- `public/data/courses.json`: **1 524 809 B (1.52 MB) / 2077 records / 16 fields**, verified by `python3 -c 'json.load(...)'`. Field set: `code`, `academy`, `name`, `englishName`, `property`, `level`, `discipline`, `hoursCredits`, `capacity`, `selected`, `teachingMode`, `examMode`, `leadProfessor`, `teachers`, `campus`, `sessions`.
- **All 2077 records carry the volatile fields** `capacity` and `selected`. Measured distributions: `selected` takes only 4 distinct values `{0, 1, 2, 118}`; 10 courses have `selected > 0`; the three at 118 are the public-required courses (`180213010108MB001H-14` 自然辩证法概论 118/310, `180213030500MB001H-14` 新时代中国特色社会主义思想理论与实践 118/245, `180096120400PB001H-05` 学术道德与学术写作规范-通论 118/300). `capacity == 0` (i.e. **not recorded**) for **99** courses.
- The app reads it from a path with no version/date: `app/page.tsx:563 fetch('/data/courses.json')`, then `app/page.tsx:566 normalizeCourse`. There is no query string, no `If-None-Match`, no timestamp assertion anywhere in the page.
- `app/page.tsx:69 capacity?: number`; is the **only** occurrence of `capacity` in the whole file (verified by `grep -n capacity app/page.tsx` → single hit). The `selected` field is likewise never read. Consequence: the app itself never renders remaining seats, so it cannot display a stale seat count —but a downstream consumer of the same file that does the natural `remaining = capacity - selected` will (a) see `0 - 0 = 0`, i.e. “**full**”, for the 99 courses that actually mean “not recorded”, and (b) see the 118-seat courses as barely-touched once enrolment moves.
- Same-origin risk already documented for the sibling data source: A1 brief, `doc/`

assignments/ a1/ A1-brief-en-2026-09-28.md:181 —” Snapshots go stale: the enrolled numbers crawled on 9-05 and those crawled on 9-15 are different (10 days apart). The system cannot know by itself that the data is stale, only by stating the capture time in the output” , and :190 —” Treating an old snapshot as 'current': it would treat an already-full section as having places left.” A1 froze its S2 snapshot on **2026-09-27**; this file is the same class of snapshot, committed without that discipline. public/ data/ README.md:23 only says ” please verify accuracy and redistribution rights before publishing or updating.”

Blast radius. Every course-derived conclusion —time conflicts, credit totals, degree-eligibility, campus/semester filtering —is computed from an unversioned snapshot. Two failure modes: (i) **staleness** —the 2027 catalogue is a different file with no signal that the old one is out of date; (ii) **semantic** —treating capacity == 0 as ” full” instead of ” unknown” (99/2077 ≈ 4.8 % of records).

Existing mitigation. public/ data/ README.md:21 documents the schema and README.md:135 warns to re-validate with official data before a new academic year. **No mitigation** for capture time, hash, or a freshness assertion in code.

Judgement: Latent. Nothing in the recorded run is wrong today (0 courses are actually full by the data: capacity>0 && capacity-selected<=0 → 0 records), but the file is already stale relative to any live term and the capacity == 0 ambiguity is present now.

R4 —Persistence is localStorage-only with no version-migration path

Phenomenon. The user's whole plan lives in one localStorage key. On read, the app accepts exactly one schema version; anything else —a future version, a truncated value, or malformed JSON —makes it **silently discard the entire saved plan**, with no message and no backup.

Evidence.

- app/page.tsx:112 const PLAN_STORAGE_KEY = 'ucas-graduate-course-planner-v3'; —note the **-v3 in the key name while the accepted payload version is 1** (below); the two version numbers do not move together.
- Read path: app/page.tsx:567 const stored = window.localStorage.getItem(PLAN_STORAGE_KEY); ...:578 if (parsed.storageVersion === 1) { ... accept ... }.
- Reject path: app/page.tsx:587–593 the else branch runs setPlan([]) and clears degreeCodes, englishPlan, studentType, selectedDiscipline —**a full silent wipe with no setMessage**. The catch at :594–600 is a byte-for-byte repeat of the same five clears, again with no message.
- Write path: app/page.tsx:619–631 writes { storageVersion: 1, plan, degreeCodes, englishPlan, studentType, selectedDiscipline }. Only version 1 is ever produced.
- Other touch points: app/page.tsx:854 and :1524 both window.localStorage.removeItem(PLAN_STORAGE_KEY); (explicit ” clear” actions —also destructive and unversioned).

- There is no export-side migration reader: the only version constant in the file is the === 1 literal at :578 (grep storageVersion → lines 571, 578, 624 only).

Blast radius. A single future change to the payload shape (e.g. bumping to storageVersion: 2, or the v3→v4 key rename the key name already invites) makes **every** returning user's saved plan vanish on their next page load, silently —they will see an empty planner and a success-looking first-run screen. Malformed/partially-written localStorage (quota, a half-finished tab close) hits the same path.

Existing mitigation. The app offers a **manual JSON download/restore** (README.md:103 “ 方案仅保存在浏览器本机， 可下载 JSON 备份或恢复备份” ; restore handler around app/page.tsx:905–913), so a user who has exported can recover. There is **no automatic** backup, no migration, no warning on reset, and no try that preserves the raw string before overwriting.

Judgement: Latent for data loss, Occurred for the silent-reset code path. The wipe code already exists and is reachable today by any non-1 value; it does not fire under the recorded happy path only because the app writes nothing but version 1.

R5 —Host environment state decides whether a “ clean” build is possible

Phenomenon. A naive npm install on this machine **succeeds** (exit 0) while silently skipping every devDependency, which then makes npm run build fail with a module-not-found on a config import. The failure looks like a broken repository but is entirely a host-level npm setting —and it cannot be seen from inside the repo.

Evidence.

- 01-npm-install.log:3,7 command npm install --registry=... --no-audit --no-fund → added 513 packages in 33s, :8 EXIT=0. Success, no warning.
- 02-build.log:12,20,28 three [UNRESOLVED_IMPORT] on vite.config.ts (@openai/sites-vite-plugin, @tailwindcss/postcss, @cloudflare/vite-plugin), then :36–47 failed to load config ... Error [ERR_MODULE_NOT_FOUND]: Cannot find package '@openai/sites-vite-plugin', :52 EXIT=1, :49 real 0m3.840s.
- 03-install-plugin.log:2,6 npm install --save-dev @openai/sites-vite-plugin@0.2.0 → up to date in 4s —a **no-op**: the package was already in package-lock.json, it had merely been omitted from the tree, which is why the “ obvious” fix changes nothing.
- 04-npm-ci.log:4–5 records the cause verbatim: 前置： 从冻结克隆件恢复原始 package-lock.json, 删除 node_modules / 环境陷阱： 本机 npm config omit=dev (故显式 --include=dev) ; the corrected install adds **560** packages (:8) vs the naive **513** (01-npm-install.log:7) —a 47-package gap, all dev.
- 05-build.log:53 after the fix: real 0m15.526s, :56 EXIT=0. Same repo, same commit —the only change was host npm config plus restoring the frozen lock file.

- `package.json:39` pins the needed package as `"@openai/sites-vite-plugin": "^0.2.0"` (a devDependency) while `vite.config.ts:1` imports it unconditionally —so dev deps are load-bearing for the build, yet an `omit=dev` host removes them invisibly.

Blast radius. Reproducibility of the build depends on a setting stored **outside the repository** (`npm config omit`). Any consumer —a grader, CI, a teammate —on a host with `omit=dev` gets a false “this repo is broken” signal, and the on-disk `node_modules` differs by 47 packages with no in-repo indicator.

Existing mitigation. The frozen `package-lock.json` (351 463 B) plus the recorded `npm ci --include=dev` command (`04-npm-ci.log:3`) fully removes the trap once known; `doc/assignments/a3/reproduction-guide.md §4` documents it. **No mitigation inside the repo** —nothing in `package.json/README/CI` guards against a host `omit=dev` (there is no CI at all, see R1).

Judgement: Occurred (it happened on the first recorded run, `02-build.log:52 EXIT=1`).

Traceability index

Risk	Files / lines	Logs
R1	<code>vite.config.ts:1,5,7-8,10,18,23,27,45,52-58</code> ; <code>.openai/hosting.json</code> ; <code>package.json:34,37,39,40,52</code> ; <code>.github/ISSUE_TEMPLATE/*</code>	<code>05-build.log:41-50</code> , <code>07-start.log:10-12</code>
R2	<code>app/page.tsx</code> (1879 lines/63 419 B), :3- 21,69,208,384,388,403,416,427,438,447,464,661- 668,774,835; <code>lib/utils.ts</code> (169 B); <code>hooks/use-mobile.ts</code> (585 B); <code>package.json:8-14</code> ; <code>components/ui/button-group.tsx:5</code>	<code>06-lint.log:9-10</code> , <code>08-e2e.log:3</code>
R3	<code>public/data/courses.json</code> (1 524 809 B/2077/16 fields, capacity,selected); <code>app/page.tsx:69,563,566</code> ; <code>public/data/README.md:21,23</code> ; <code>README.md:135</code> ; <code>A1-brief-en-2026-09-28.md:181,190</code>	—(static-file analysis)
R4	<code>app/page.tsx:112,567,570-578,587-600,619-631,854,1524</code>	—(static-file analysis)

Risk	Files / lines	Logs
R5	<code>package.json:39; vite.config.ts:1</code>	<code>01-npm-install.log:3,7-8;</code> <code>02-build.log:12,20,28,36-47,52;</code> <code>03-install-plugin.log:6;</code> <code>04-npm-ci.log:4-5,8;</code> <code>05-build.log:53,56</code>

Correction carried forward: one claim in A1 is not supported by the code

What A1 says. `doc/assignments/a1/A1-brief-en-2026-09-28.md:123` lists what this tool already does: " campus filtering by the 18th digit of the code, **week-level conflict detection, capacity and number enrolled**, and degree-course credits with the " 2 + 2" structure check" .

What the frozen commit actually does. Campus filtering, week-level conflict detection and the 2+2 check are all confirmed here (R1/R2 evidence, `architecture-map.md §7`, and the executed end-to-end run). **Capacity and number enrolled are not a feature of the application:** `capacity` appears in `app/page.tsx` exactly once, as a type declaration (`:69, capacity?: number`), and is never read; `selected` never appears as a data field at all. The only traces of the idea are two pieces of static advice text (" 分班课名额有限…" :1359, " 课程名额通常有限" :1366) and one filter labelled " 只看当前课表空闲课程" (:1644), which consults the chosen plan, not enrolment numbers. The two fields exist in `public/data/courses.json` and are simply unused.

Why this is recorded here instead of edited there. A1 was submitted inside its own seven-day window and is frozen at tag `a1-v2`; rewriting submitted work would break the provenance this project is built on. The correction therefore lives in A3, and is carried forward to A5 (quality and counter-examples), where " a claim in our own earlier deliverable that the evidence does not support" is exactly the kind of counter-example the course asks for.

How it changes the argument. A1's conclusion —that this tool cannot answer " can this course wait until spring" , has no order to register in, and carries no provenance for its data —is unaffected. What changes is one item in the list of features it does have: it shows clash information and a 2+2 credit check, but it never shows how full a section is. If anything, that strengthens A1's gap ③: the tool carries enrolment fields it never surfaces, which is precisely the " static, sourceless snapshot" problem A1 names.

\newpage

ADR-001 —Establishing a safe change boundary: two responses

- **Status:** Accepted

- **Date:** 2026-10-05
 - **Deciders:** A3 investigation (author)
 - **Subject repository:** third-party UCAS-course-planner, commit 68e7080503084a41f42db6e0e9ff (ver 1.4, 2026-09-01), investigated read-only
 - **Risk addressed:** R2 in doc/assignments/a3/risk-register.md — “One 1879-line component carries all logic; no safe change boundary”
 - **Required by:** A3 brief requirement 5, “Identify three risks; compare two responses to one in an ADR” (课程材料/ISE-A3_Repository_Investigation_and_Reproducible_Build_Engl line 19), with deliverable “ADR and execution evidence” (line 27) and objective “establish a safe change boundary” (line 11).
-

1. Context and constraints

The application's entire domain logic —week/period parsing, conflict detection, degree/credit rules, persistence, and every UI branch —lives in a single client component. The frozen checkout measures `app/page.tsx` at **1879 lines / 63 419 B**, while the only supporting files are `lib/utils.ts` (**169 B / 6 lines**, just the `shadcn cn()` helper) and `hooks/use-mobile.ts` (**585 B / 21 lines**). `app/page.tsx:3-21` imports **only** `react` and `lucide-react`; it does not import either of those files (they are referenced only by unused components/ui/* scaffolding, e.g. `components/ui/button-group.tsx:5`). The file declares **41 functions** —**34 at top level** plus **7 mutation handlers nested inside the component** (`:809, :825, :835, :845, :853, :864, :891`) —including the whole domain core: `parseWeeks (:403)`, `parsePeriods (:416)`, `intersects (:427)`, `hasSlot (:438)`, `coursesConflict (:447)`, `conflictsWithPlan (:464)`, `isCoreCourse (:384)`, `isProfessionalCourse (:388)`.

Two hard constraints from the evidence shape this decision:

- **C1 —there is no automated correctness check to lean on.** `package.json:8-14` defines `dev / build / start / lint / format` and **no test script**. The linter inspects only **5 files** (`06-lint.log:10: Finished in 2.6s on 5 files with 208 rules using 8 threads.`) and reports the monolith clean (`06-lint.log:9: Found 0 warnings and 0 errors.`), which carries no information about behaviour. The only recorded end-to-end check is a manual `localStorage` injection followed by a re-render (`08-e2e.log:3`), and it needs the full build + a live server.
- **C2 —every verification run is expensive and platform-bound.** A build costs `real 0m15.526s` (`05-build.log:53`) after neutralising the host `omit=dev` trap and restoring the frozen lock (`04-npm-ci.log:4-5`), because the build chain is coupled to a hosting vendor (R1: `vite.config.ts:1,5,52-58`). A unit-level check would need neither the vendor chain nor a server.

The follow-on assignment (A4) requires making a **bounded change to this codebase** —“cutting one slice”. The decision below must therefore be defensible not only for a one-off fix but for a change that will be reviewed, and for the case where the change is later extended

or reverted.

Because the repository is third-party and read-only for A3, this ADR **selects a boundary for a future change** (A4); it does not itself modify UCAS-course-planner. The two options differ in where the change is made and what can verify it, not in feature scope.

2. Options

Option A —Patch inside the monolith (minimal diff)

Make the change directly in `app/page.tsx`, editing the relevant functions in place; add no new files and no test harness.

Benefits

- **Smallest possible diff and no new API surface.** The functions already sit at module scope sharing existing helpers (`coursesConflict` at :447 is already called from two places, :468 and :665), so a targeted edit is local.
- **No new maintenance surface.** It introduces no `lib/` module, no build/test tooling, and no import boundary the repo must then keep consistent —relevant because the repo has **no CI** (`.github/` holds only `ISSUE_TEMPLATE/*`) to enforce consistency automatically.
- **Fastest to land.** Nothing has to be installed or configured; the existing `npm run dev / build` path already works (`05-build.log:56 EXIT=0`).
- **Right answer if the change is genuinely one-off and short-lived** (e.g. a text/label fix, or a probe that will be reverted): extraction would be pure overhead.

Costs

- **Unverifiable.** After the edit there is still no test script and no per-rule check; correctness rests on a manual `localStorage` E2E run that requires the whole platform build chain (C1, C2). A regression in `coursesConflict` would surface only as a wrong conflict badge in a live browser.
- **It re-creates the risk it is supposed to answer.** R2's finding is precisely that no boundary exists; editing in place entrenches it, and a second caller cannot exercise the changed rule in isolation.
- **Review cost scales with file size.** Reviewers must hold a 1879-line/63 KB file plus 21 `useState` and 14 `useMemo` sites in mind to judge a diff that touches a shared domain function (`useState` 21, `useMemo` 14, 冲突 9, 学位课 19 occurrences / 17 lines over `app/page.tsx`).

Option B —Extract a pure, independently testable logic layer

Move the **pure, DOM-free** rules out of `app/page.tsx` into `lib/` (conflict judgement: `parseWeeks`, `parsePeriods`, `intersects`, `hasSlot`, `coursesConflict`, `conflictsWithPlan`; degree/ credit judgement: `creditFromValue`, `isCoreCourse`, `isProfessionalCourse`,

matchesDiscipline and the credit predicates), add a minimal test for them, and have the page **import and call** them. The page keeps all React state and rendering.

Benefits

- **Creates the missing verification seam.** A pure function can be checked without React, without the vendor build chain, and without a server —directly resolving C1/C2. This is the only option that can demonstrate the changed rule is correct rather than hope it is.
- **Answers the A3 objective literally.** A3 line 11 asks to "establish a safe change boundary" ; B produces one, so the A4 slice lands on a narrow, testable surface instead of inside the monolith. A reviewer of A4 can see the diff at function granularity plus a passing test.
- **Cheap to do here because the functions are already pure leaves.** The chosen functions take only data (Course, Set<number>, string) and return booleans/numbers —no hooks, no JSX, no window —so the move is mostly cut-and-paste plus an **export**, which is why the extraction is bounded and low-risk rather than a rewrite.
- **Blast radius of a wrong extraction is caught immediately** by the test, before the page is touched.

Costs

- **New surface and new tooling.** It adds `lib/*.ts` plus a test file, and the repo currently has **no test runner at all** (`package.json:8–14`) —so a minimal runner must be added and justified, and it becomes something the repo must maintain even though there is no CI to run it.
- **A visible, reviewable diff instead of a hidden one.** The change to `app/page.tsx` is larger in line count (it now imports and delegates) even though each function's body is unchanged.
- **Up-front effort for a possibly tiny change.** If the A4 slice turns out to be UI-only (e.g. relabelling a status string near `app/page.tsx:1165–1168`), the seam is never used and the effort is wasted.
- **Must not touch persistence.** Any extraction that reaches into the `localStorage` read/write path (`:567, 621`) risks colliding with R4's silent-reset behaviour, so the extraction must be strictly limited to pure rules.

3. Decision

Choose Option B, with a deliberately bounded scope: extract **only the pure, DOM-free rule functions** listed above, add **one minimal test** that exercises `coursesConflict / conflictsWithPlan` against a positive and a negative case, and leave **all** React state, effects, rendering, and persistence exactly where they are.

Reasons, and why this holds under A4's "cut one slice" :

1. **The decision must be defensible to a reviewer, and only B produces verifiable evidence.** C1 shows the repo has no test script and a linter that sees 5 files and already passes on the monolith. Under A the claim " the change is correct" is unfalsifiable except by a full platform build plus a manual browser check; under B it is backed by a green test that needs neither. A3's rubric weights " Execution, artifact quality, and reproducibility" (1.50) and " Analysis, counterexample, and limits" (1.25) above convenience (ISE-A3...English.md lines 43-44), and the brief demands " Establish a safe change boundary" (line 11) —A reproduces the condition R2 names; B removes it.
2. **The A4 slice lands on the boundary, not on the monolith.** Whatever slice A4 cuts, a reviewer/automation can act on it at function granularity, and the pure-rule test stays green as a guard against regressions in conflict/credit behaviour —the exact logic A1 identified as the place prior errors occurred (A1-brief-en-2026-09-28.md:60: the three courses all in " Thursday, periods 10-12" ; :201-207 the rejected advice to stop cross-checking conflict inputs).
3. **The dominant benefit is realised even if the slice is small, because the seam is permanently cheaper than the E2E alternative.** Every future conflict/credit check stops costing a ~15.5 s build plus a server (05-build.log:53, 07-start.log:12).
4. **B is bounded and reversible for a concrete reason, not optimism:** the target functions are already pure leaves with a single existing call pattern (coursesConflict already called from :468 and :665), so the extraction does not entangle state —which is what makes the cost estimate credible rather than a rewrite in disguise.

Explicitly rejected scoping that B must NOT be interpreted as authorising: extracting hooks, effects, or the persistence path (that would collide with R4), or " refactoring the page" wholesale —the decision is narrow.

4. Cost of the rejected option (Option A)

Option A is not wrong in general; it is wrong **here**. Its real costs, stated honestly, are:

- **It cannot produce the evidence A3/A4 are graded on.** With no test script and a linter that inspects 5 files and already passes on the untested monolith, an in-place patch leaves the correctness claim resting on a manual `localStorage` render (08-e2e.log:3) that requires the coupled vendor build chain (R1).
- **It converts a one-time fix into a permanent constraint.** A future caller cannot test the changed rule in isolation, so every subsequent change to conflict/credit logic pays the same verification tax —the reasoning that makes the first in-place edit cheap is exactly what makes the next one expensive.
- **What A buys (no new files, no new tooling) is real and partially offsets B's cost.** If the A4 slice were confirmed to be UI-text-only, A would win on effort; the decision therefore also records when to prefer it (see § 6).

No option here is free: A pays in unverifiability and entrenched risk; B pays in added tooling

and a wider (if shallow) diff.

5. Rollback

Option B is rolled back in **one commit** with **no data-path change**, because the extraction does not touch state or persistence:

1. Re-inline the extracted functions into `app/page.tsx` (their bodies are unchanged from the pre-extraction versions in the frozen checkout at :403–470 and :384–401), restoring the original imports.
2. Delete `lib/<extracted>.ts` and the added test file; remove the temporary test runner added to `package.json:8–14`.
3. Re-run `npm run build + npm run lint —expected EXIT=0 and Found 0 warnings and 0 errors. (05–build.log:56, 06–lint.log:9)`, i.e. back to the frozen baseline.

Because the plan/persistence contract (`app/page.tsx:112, 567–631`) is untouched, a rollback **cannot** trigger R4's silent reset for any user. That is a deliberate property of the chosen scope, not a side effect.

6. Revisit trigger

Reopen this decision if **any** of the following becomes true:

- **The slice requires state, effects, or persistence, not pure rules.** If A4's change touches `useState/useEffect` (21/3 occurrences) or the `localStorage` path (:112,567,621), the bounded extraction no longer applies and Option A (or a different decomposition) must be re-evaluated —the seam would otherwise be a fiction.
 - **The extraction touches the persistence contract at all.** That is a R4 hazard; stop and split the change.
 - **The A4 slice is confirmed UI-only** (e.g. a label near :1165–1168), in which case the pure-rule seam is unused and A is the cheaper correct answer.
 - **The repository gains a test runner and CI such that in-place edits become independently verifiable.** Then A's principal cost disappears and the balance may shift; note the repo currently has no CI (`.github/` contains only `ISSUE_TEMPLATE/*`).
 - **A host/build-portability change (R1) or a data-freshness change (R3) is the actual slice**, since neither is answerable by this seam and each needs its own decision record.
-

Traceability

Statement	Source
R2 as stated	doc/assignments/a3/risk-register.md, R2
A3 objective / requirement 5 / rubric weighting	课程材料/ISE- A3_Repository_Investigation_and_Reproducible_B lines 11, 19, 27, 43-44
Monolith size, imports, function inventory	app/page.tsx (1879 lines / 63 419 B); :3-21, 208, 384, 388, 403, 416, 427, 438, 447, 464, 468, 665
Supporting-file sizes; unused by the page	lib/utils.ts (169 B); hooks/ use-mobile.ts (585 B); components/ui/ button-group.tsx:5
No test script; scripts list	package.json:8-14
Lint reach	06-lint.log:9-10
E2E method	08-e2e.log:3
Build cost / success	05-build.log:53, 56; 07-start.log:12
Host omit=dev trap and lock restore	04-npm-ci.log:4-5
Vendor coupling (R1)	vite.config.ts:1, 5, 52-58
Persistence contract and reset path (R4)	app/page.tsx:112, 567, 578, 587-600, 619-631
Prior conflict-error precedent	doc/assignments/a1/ A1-brief-en-2026-09-28.md:60, 201-207
No CI	.github/ISSUE_TEMPLATE/* (only files present)

\newpage

Appendix —diagrams and the executed run

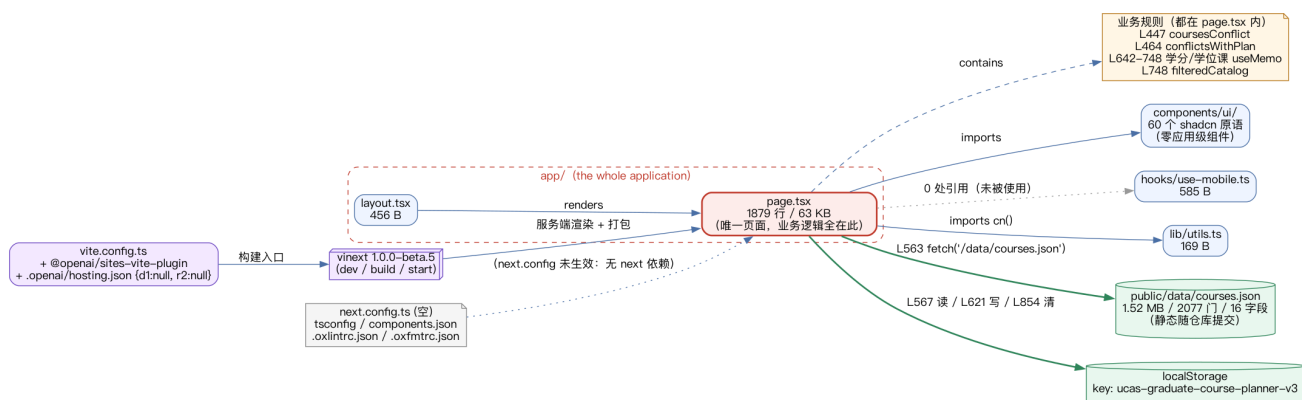


Figure A3-1 —Module map of the frozen checkout.

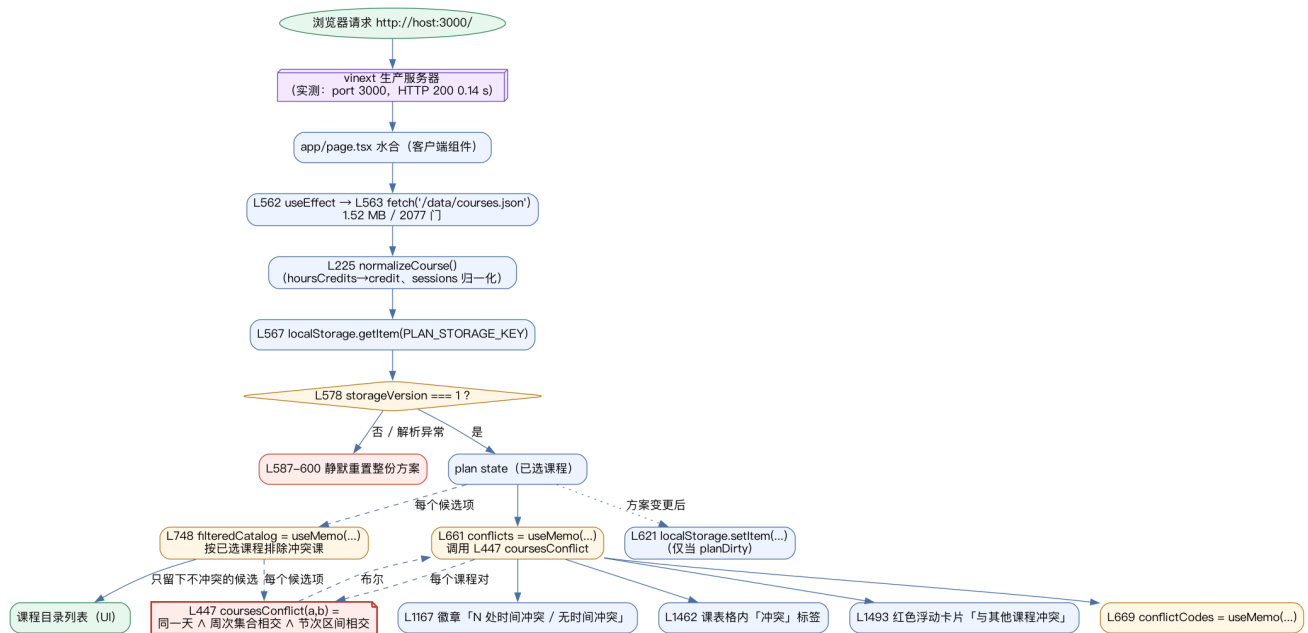


Figure A3-2 —Call path of one end-to-end run, with line numbers.

	14:20–15:05				
7	15:25–16:10				
8	16:15–17:00				
9	17:05–17:50				
10	18:30–19:15				冲突 计算机代数在科学与工程中的应用 教学楼304 · 1分 与其他课程冲突 深度学习方法与应用 教学楼阶一6 · 2分 与其他课程冲突 数字图像处理与分析 教学楼212 · 2.5分 与其他课程冲突

Figure A3-3 —Executed positive case: three courses sharing 周四 10-12 → 「3 处时间冲突」 and a red 「冲突」 badge in the period-10 row.

2026 秋季 · 博士生课程规划

博士选课规划

本地优先 · 从空方案开始

将培养要求、时间冲突、考核方式和课程数据放在同一个工作
面里，先规划，再去本校选课系统提交。

● 规划器已就绪

秋季方案

4.0 本学期已选学分

2 门课程

1 门闭卷

秋季 ≥10 分

春季 ≥10 分

夏季 ≥2 分

核心课 ≥2 门

学位课 ≥16 分

总学分 ≥38 分

个性化规则检查

先设置身份与一级学科

仅保存在本机

研究生身份

本人一级学科

Figure A3-4 —Executed negative control: no shared slot → 「无时间冲突」.